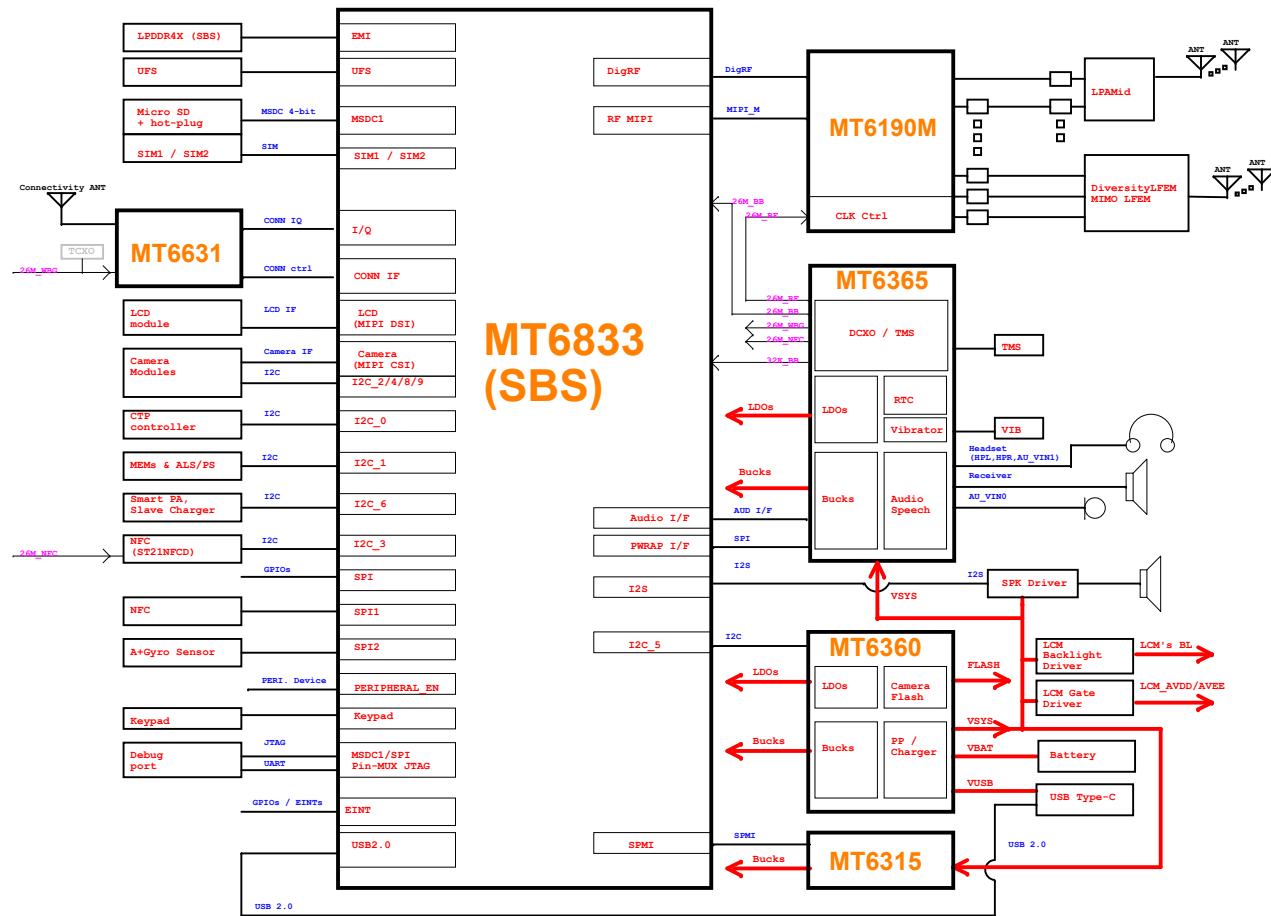
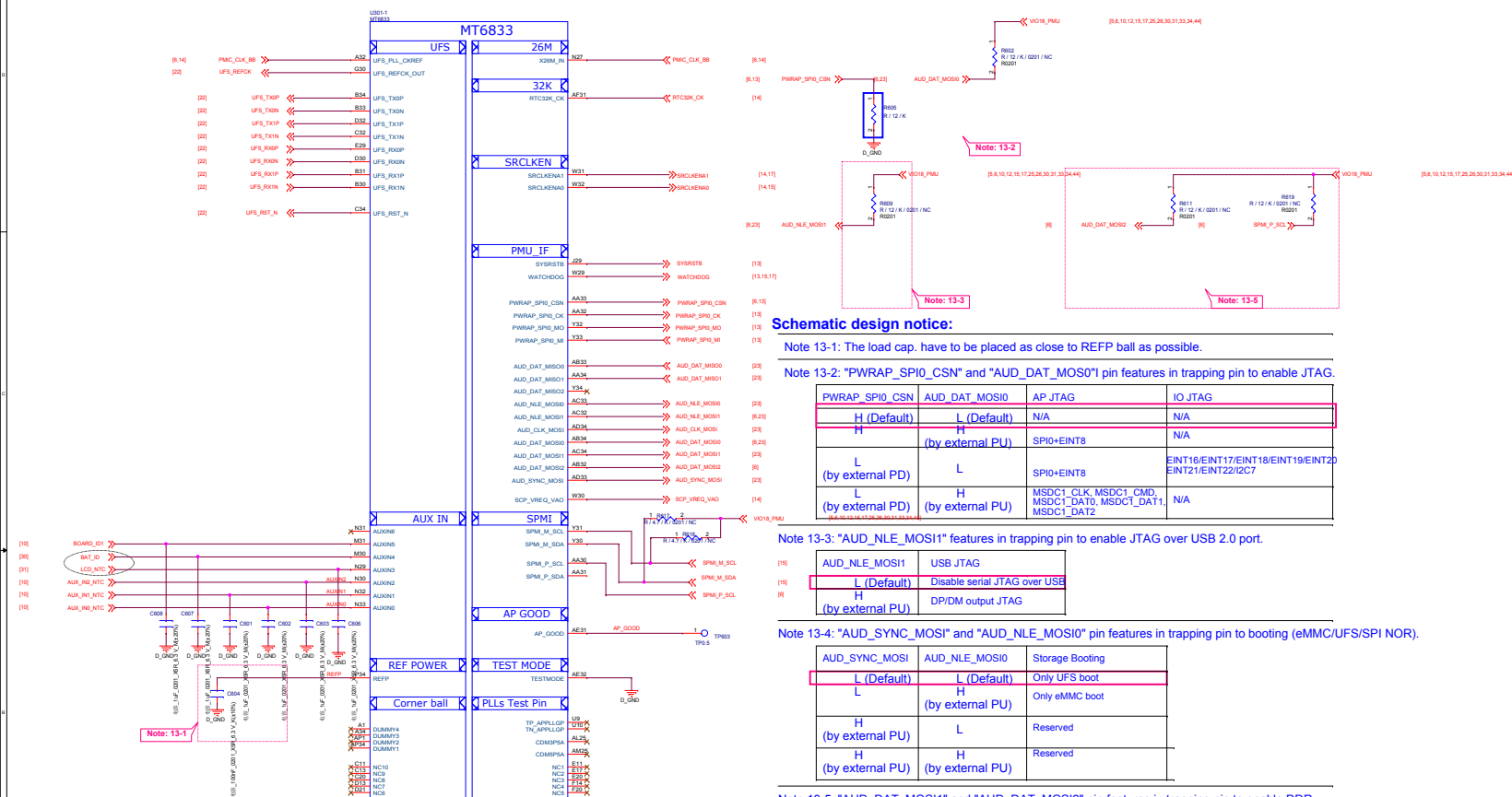




华勤通讯 Huaqin Telecom Technology Com., Ltd		
Title Block Diagram		
Size D	Project <Doc>	Rev V1
Date: Tuesday, January 12, 2021	Sheet 1 of 48	

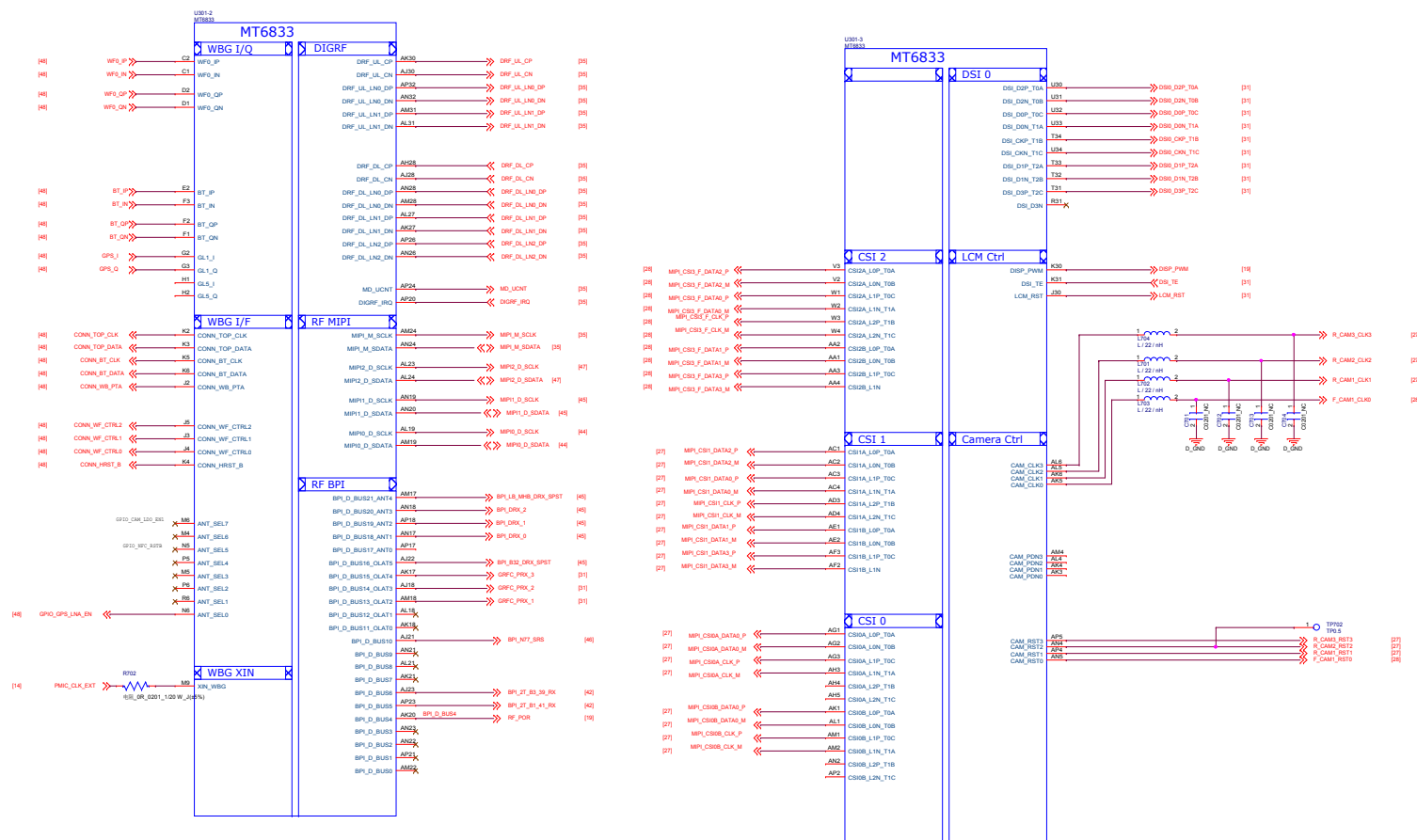
Date	Version	Page	Change notice
2020.06.xx	Pre-Released		
2020.07.xx	V01		
2020.08.14	V02	13_BB_1 29_POWER_EXT 20_POWER_MT6365_Buck 25_POWER_MT6315_VMD	Disconnect Dummy pins A1,A34,AP1,AP34 to GND Change C2923 cap from 6.3V to 10V [PDN] Change Caps of DVDD_GPU - C2039/C2038 to NC [PDN] Change Caps of DVDD_MODEM - C2502/C2534/C2523/C2522 to NC
2020.08.26	V03	File name	Modify the file name

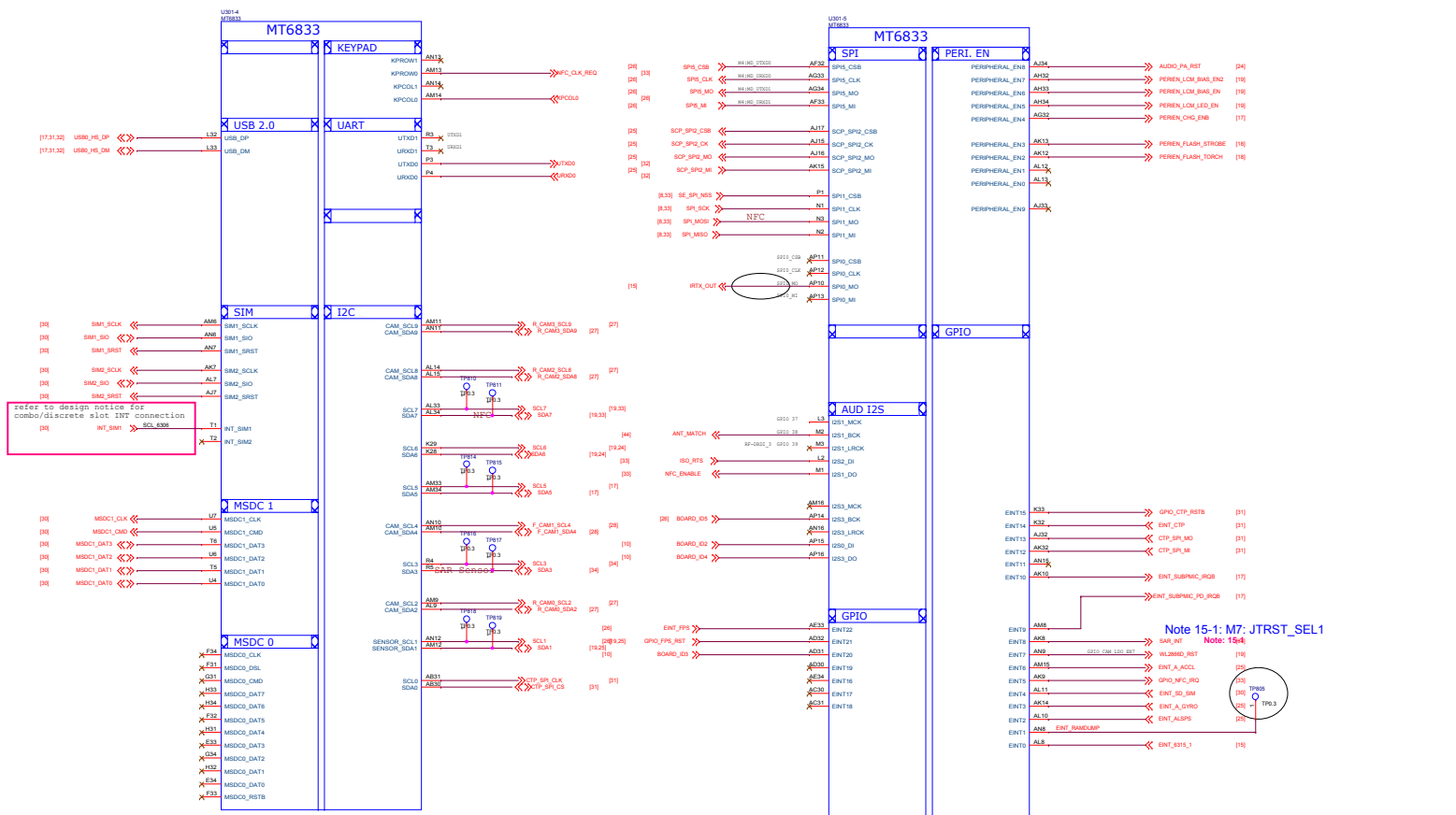
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title SM6350_Control		
Size D	Project <Doc>	Rev V1
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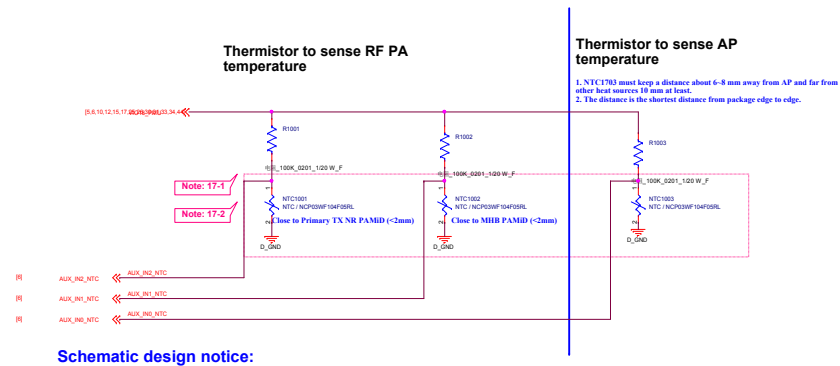
华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title			06.BB_1
Size	Project	<Doc>	
D			Rev v1
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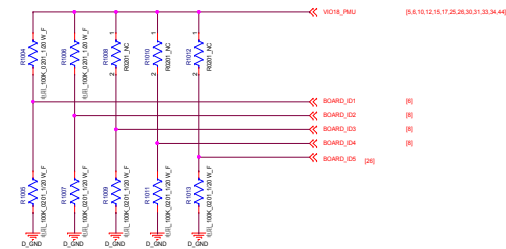
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		08.BB_3	
Size	Project	<Doc>	Rev
D			v1
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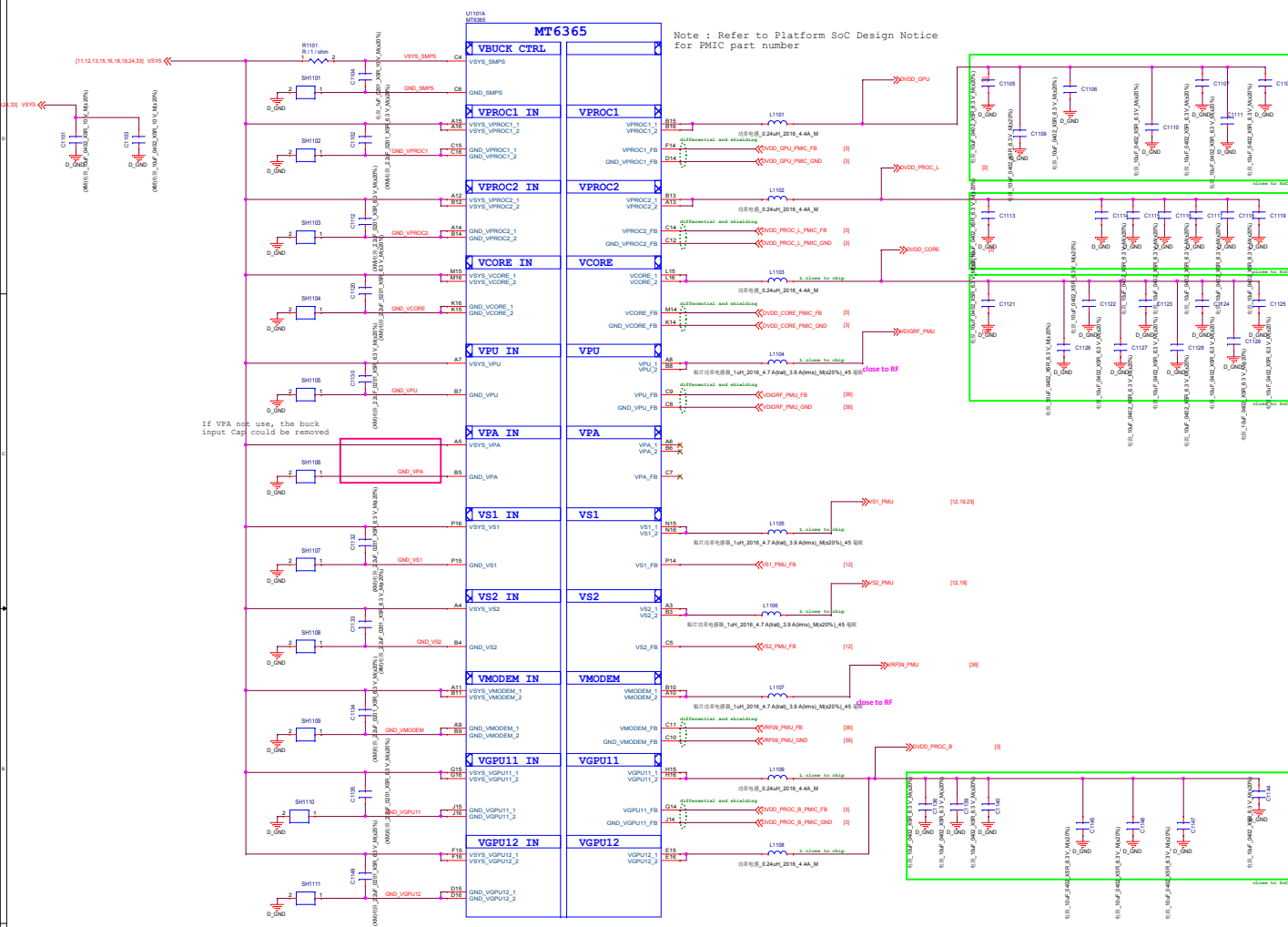
Schematic design notice:

Note 17-1: SW Default Configuration is as followings
AUX_IN0_NTC is for AP Temp.
AUX_IN1_NTC is for MHB PAMID Temp.
AUX_IN2_NTC is for NR PAMID Temp.

Note 17-2: If your design or placement is different with SW Default Config.
Please refer to "MTXXXX Thermal User Manuel.docx" and modify SW setting.



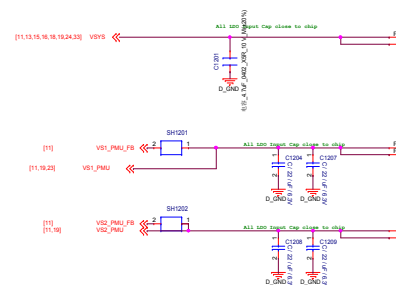
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 10.BB_AUXADC_Thermal/BOARD_ID			
Size D	Project <Doc>		Rev v1
Date: Tuesday, January 12, 2021	Sheet 10 of 48		



Circuit Type	Buck Name (Application name)	Output Voltage Range (V)	Boot Default (V)	I _{OUT-MAX} (mA)
Buck	VPROC1 (DVDD_GPU)	0.40 ~ 1.19 (6.25mV/step)	ON (0.75)	4800
	VPROC2 (DVDD_PROC_L)	0.40 ~ 1.19 (6.25mV/step)	ON (0.75)	4800
	VGPU11+12 (DVDD_PROC_B)	0.40 ~ 1.19 (6.25mV/step)	ON (0.75)	4800 *2
	VCORE (DVDD_CORE)	0.40 ~ 1.3 (6.25mV/step)	ON (0.75)	4800
	VMODEM (VRF09_PMU)	0.50 ~ 1.10 (6.25mV/step)	OFF	4800
	VPU	0.40 ~ 1.19 (12.5mV/step)	ON (0.7)	2400
	VS1 (VS1_PMU)	1.86 ~ 2.20 (12.5mV/step)	ON (2.0)	2200
	VS2 (VS2_PMU)	1.20 ~ 1.50 (12.5mV/step)	ON (1.35)	2500
	VPA (VPA_PMU)	0.50 ~ 3.40 (50mV/step)	OFF	1000

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		11.POWER_MT6365_Buck	
Size	Project	<Doc>	
D		Rev	
Date: Tuesday, January 19, 2021		Sheet 11 of 48	
		V1	

MT6365

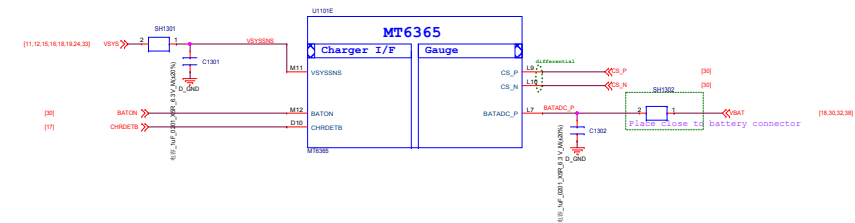


Circuit Type	LDO Name	Output Voltage (V)	Boot Default (V)	IOUTMAX (mA)	Expected use
DLD0	VSIM1	1.7/1.8/1.96/ 2.76/3.0/3.1	OFF (1.86)	200	SIM
	VSIM2	1.7/1.8/1.96/ 2.76/3.0/3.1	OFF (1.86)	200	SIM
	VCN33_1	3.3/3.4/3.5/3.6	OFF (3.3)	800	Connectivity
	VCN33_2	3.3/3.4/3.5/3.6	OFF (3.3)	800	Connectivity
	VEMC	2.55/2.9/3.0/3.3	ON (3.0)	800	eMMC / UFS
	VIO28	2.4/2.9/3.0/ 3.1/3.2/3.3	OFF (2.8)	200	IO & Sensor
	VIBR	2.7/2.8/3.3	OFF (2.8)	200	Vibrator

Title	12.POWER_MT6365_LDO
-------	---------------------

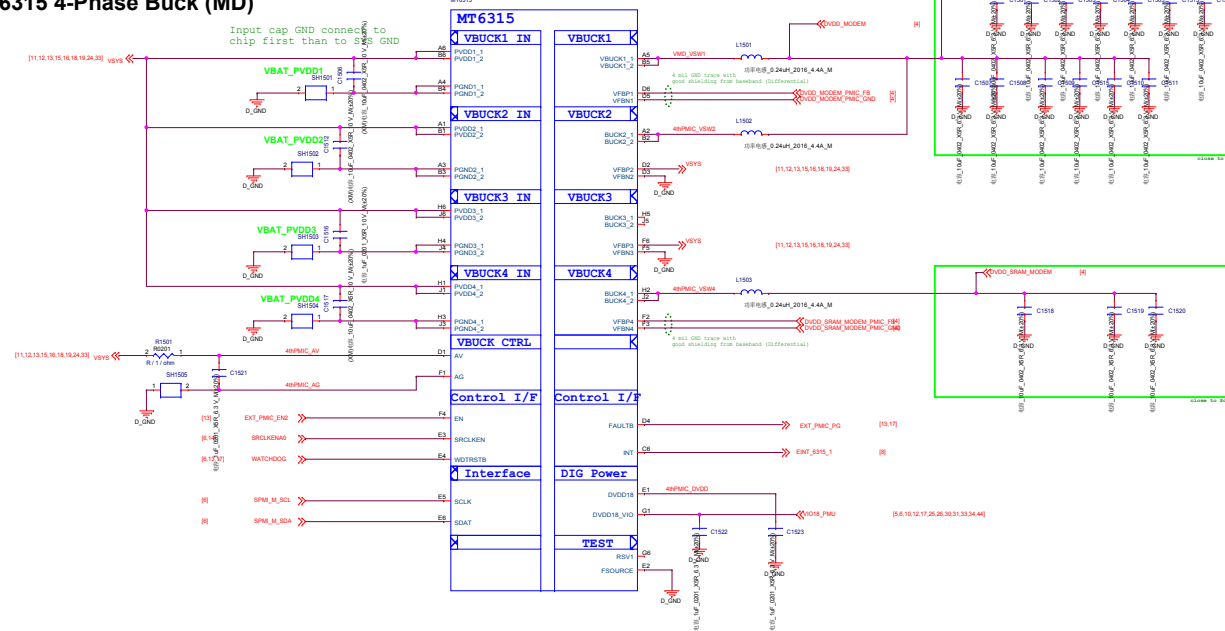
<Doc>

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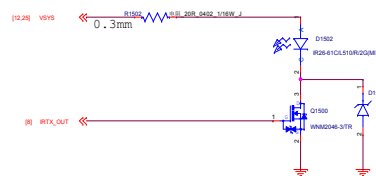
MT6315 4-Phase Buck (MD)

Note : Refer to Platform SoC Design Notice for PMIC part number

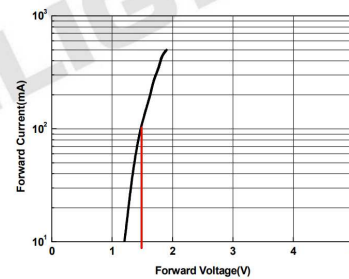


MT6315NP/B	Output Voltage Range (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
BUCK1+2	0.40 ~ 1.19375 (6.25mV/step)	ON (0.825)	5000*2	DVDD_MODEM
BUCK3	0.40 ~ 1.19375 (6.25mV/step)	ON (0.75)	5000	—
BUCK4	0.40 ~ 1.19375 (6.25mV/step)	ON (0.825)	5000	DVDD_SRAM_MODEM

IR LED



Forward Current vs. Forward Voltage



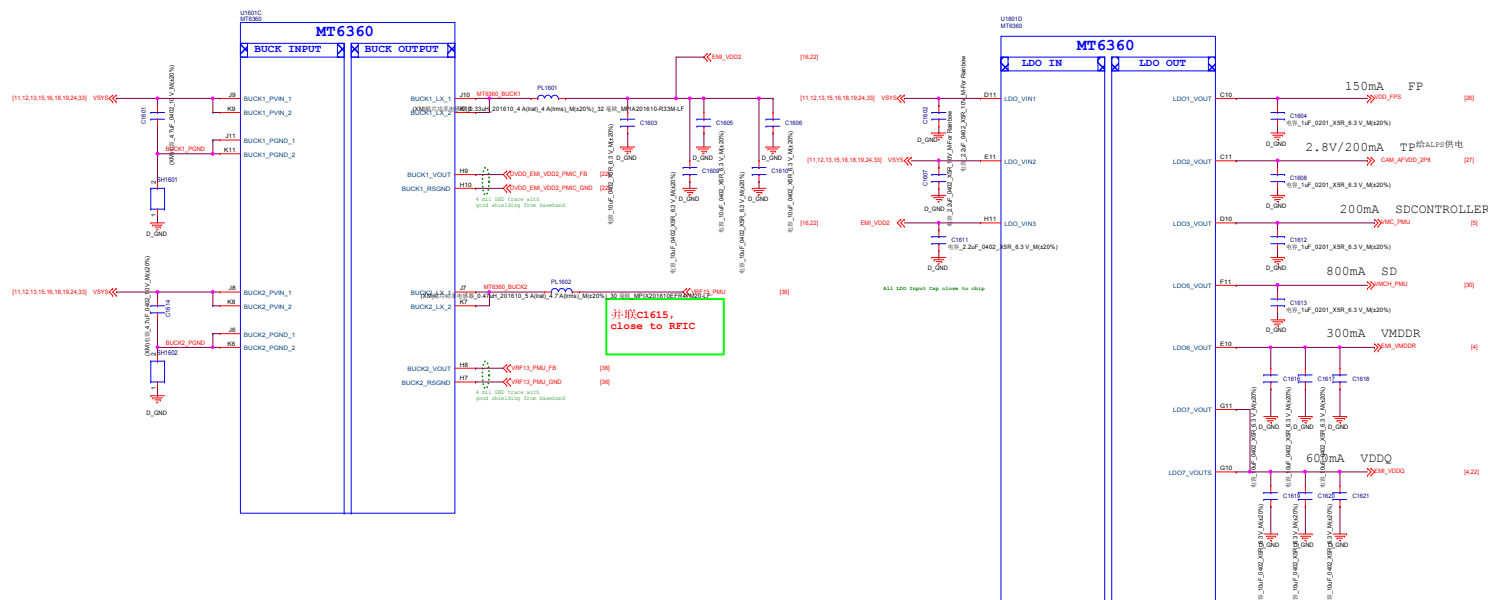
华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 15.POWER_MT6315_VMD/IR_LED

Size D Project <Doc> Rev V1

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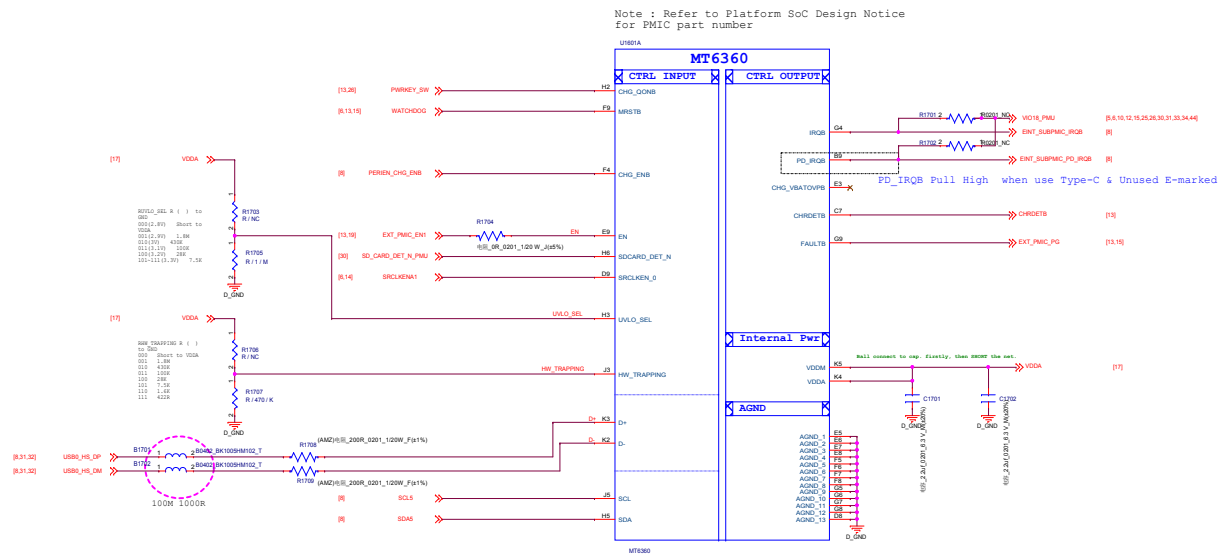
Note : Refer to Platform SoC Design Notice
for PMIC part number



Circuit Type	Name	Output Voltage Range (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
Buck	BUCK1	0.3 ~ 1.3 (5mV/step)	ON (1.125)	3000	EMI_VDD2
	BUCK2	0.3 ~ 1.3 (5mV/step)	OFF (1.3)	3000	VRF13_PMU

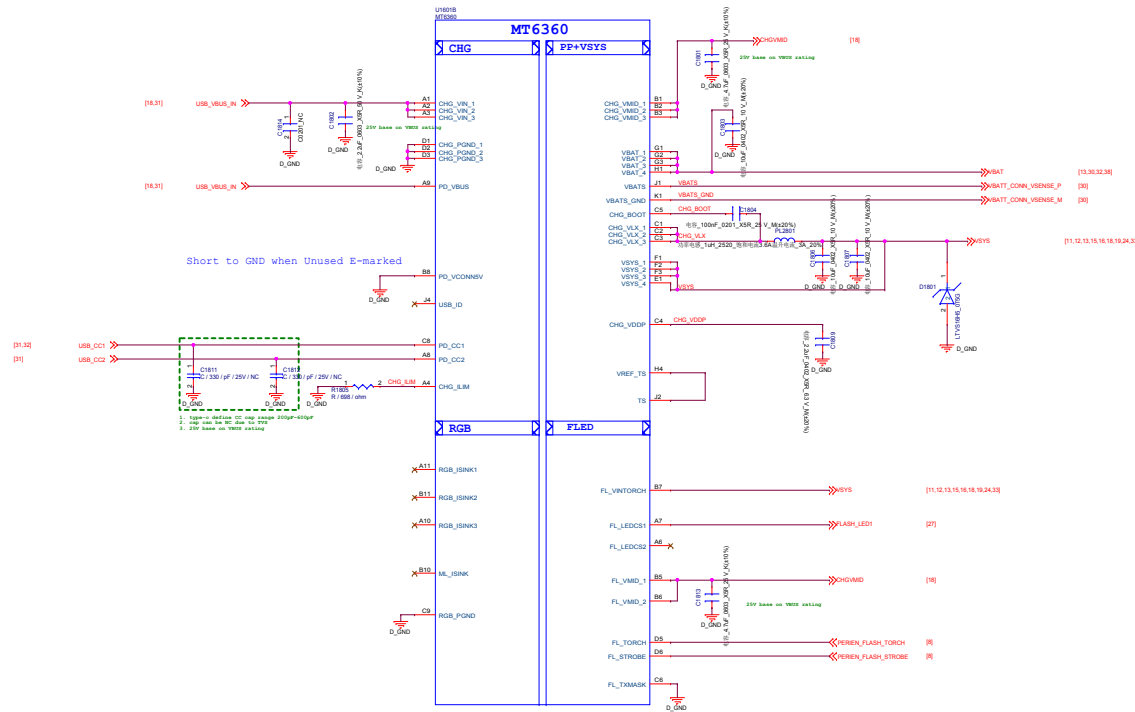
Circuit Type	Name	Output Voltage (V)	Boot Default (V)	I _{OUT-MAX} (mA)	Expected use
LDO	LDO1	1.8/2.0/2.1/2.5/2.7/ 2.8/2.9/3.0/3.1/3.3	OFF (1.8)	150	Fingerprint
	LDO2	1.8/2.0/2.1/2.5/2.7/ 2.8/2.9/3.0/3.1/3.3	OFF (1.8)	200	Touch Panel
	LDO3	1.8/2.9/2.8/3.0/3.3	OFF (3.0)	200	SD Card
	LDO5	2.9/3.0/3.3	OFF (3.0)	800	SD Card
	LDO6	0.75	ON (0.75)	300	EMI_VMDDR
	LDO7	0.6	ON (0.6)	600	EMI_VDDQ

华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 16.POWER_MT6360_BUCK_LDO*		
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华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 17.POWER_MT6360_General		
Size D	Project <Doc>	Rev v1
Date: Tuesday, January 12, 2021	Sheet 17 of 48	

Note : Refer to Platform SoC Design Notice
for PMIC part number



Schematic design notice of "28_POWER_MT6360_Charger" page.

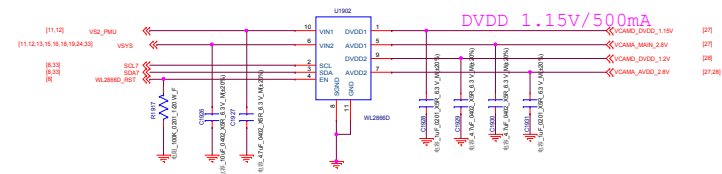
Note 28-1: For better ESD & surge performance we need choose suitable device for system protection.
Please refer to the latest version of [Surge device selection guide] provided by MTK.

华勤通讯 Huaqin Telecom Technology Com.,Ltd

Title 18.POWER_MT6360_Charger

Size D	Project <Doc>	Rev v1
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[illegible][illegible][illegible]

厂家	IPC ADDRESS
KL2866D	0a50H
KT5904	0a50H

厂家	INC ADDRESS
OC92131	0111110 binary (or 3Eh)
SMC109	0111110 binary (or 3Eh)

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title 19.POWER_EXT			
Size D	Project <Doc>		Rev V1
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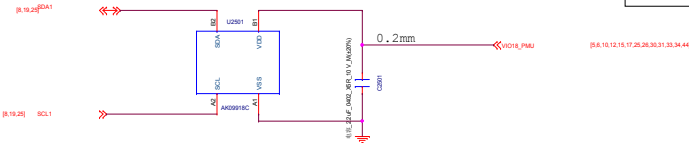
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title PM7250B_CHARGE/FG		
Size E	Project <Doc>	Rev v1
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华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title PM7250B_MISC/GPIO/GND		
Size D	Project <Doc>	Rev v1
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eCOMPASS

M-Sensor

厂家	I2C ADDRESS
AK8963C	0ch
AK8963C	2ch

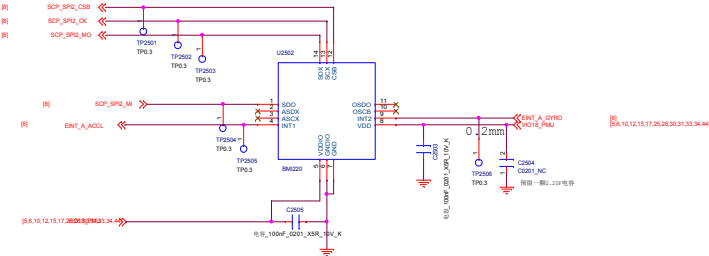


A+Gsensor

default SPI (if sensor hub support)

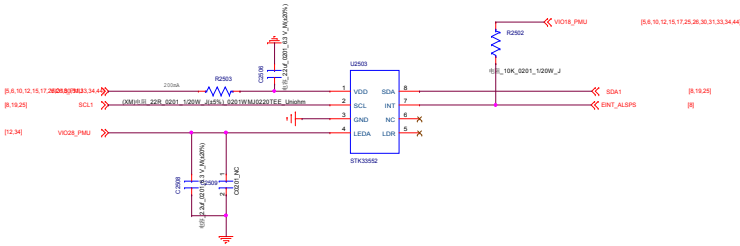
厂家	I2C ADDRESS
AK8963C	
AK8963C	

Gyro+A sensor



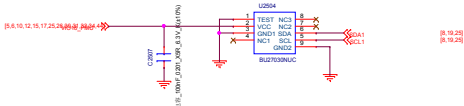
ALS+PS

厂家	I2C ADDRESS
AK8963C	0ch
AK8963C	2ch

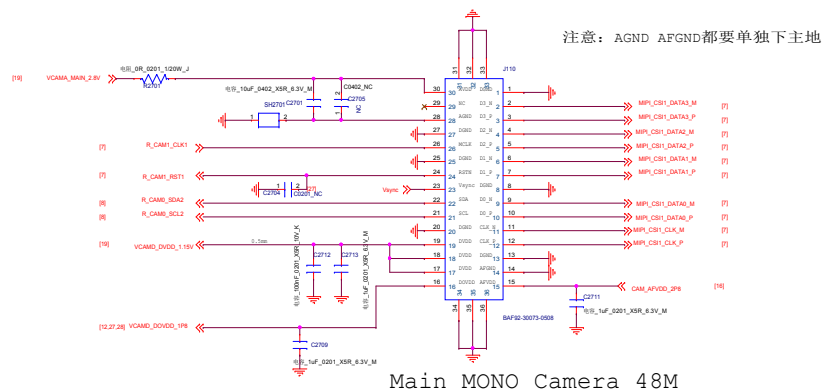


后光感 BU27030NUC

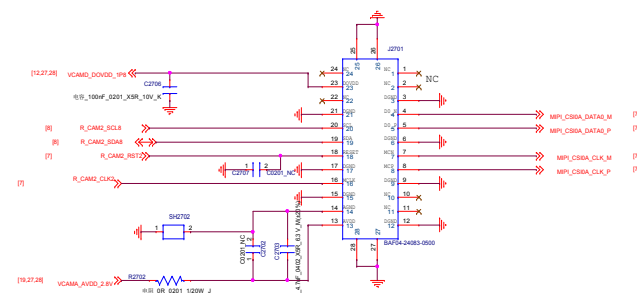
厂家	I2C ADDRESS
AK8963C	0ch
AK8963C	2ch



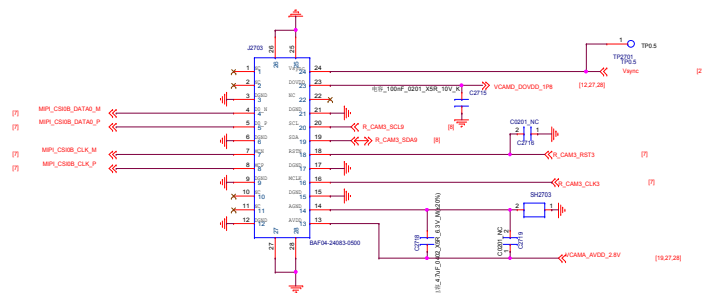
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		25.SENSOR	
Size D	Project	<Doc>	Rev v1
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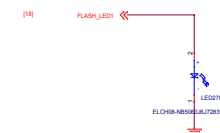
微距 Camera 2M =Macro



景深 Camera 2M=Depth



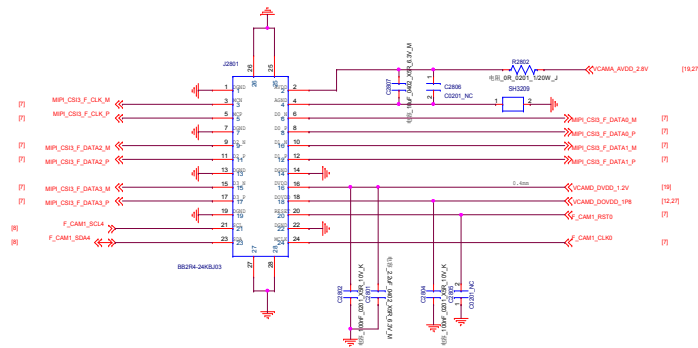
Flash LED 5V Boost



Flash 下方地要完整且足够大, 散热一定要好

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		27.REAR_CAM	
Size	Project	<Doc>	Rev
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Front Camera - 8M



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title28.FRONT_CAM			
SizeD	Project<Doc>		RevV1
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华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		Sub_Charger_SMB1396	
Size D	Project <Doc>		Rev v1
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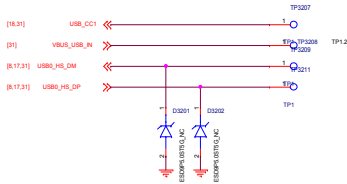
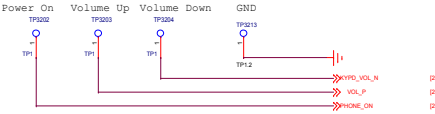
VBAT TEST POINT



VBAT

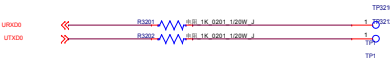
KEYPAD

KEY BTB

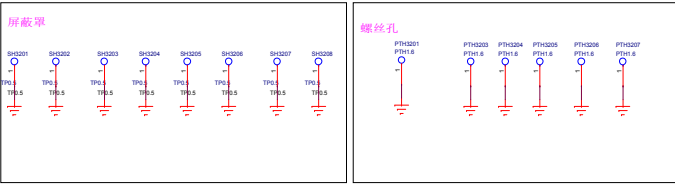


VCHG

UART TP

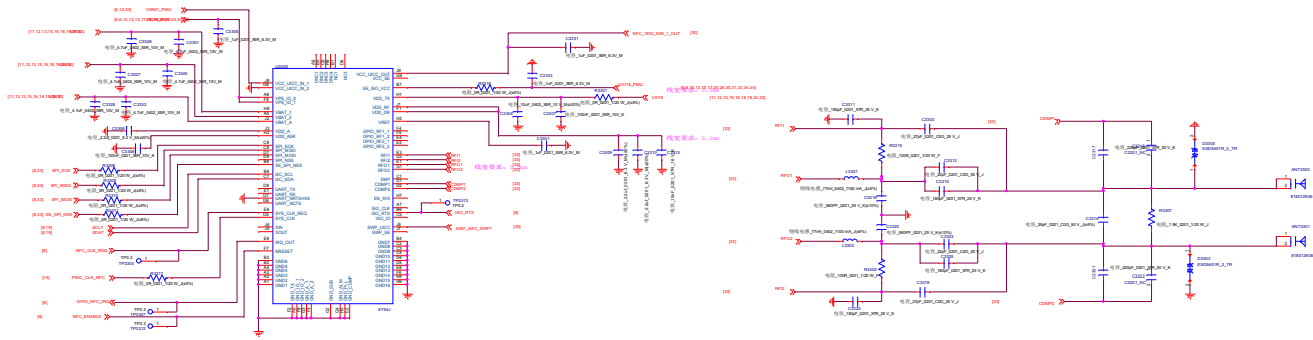


JTAG



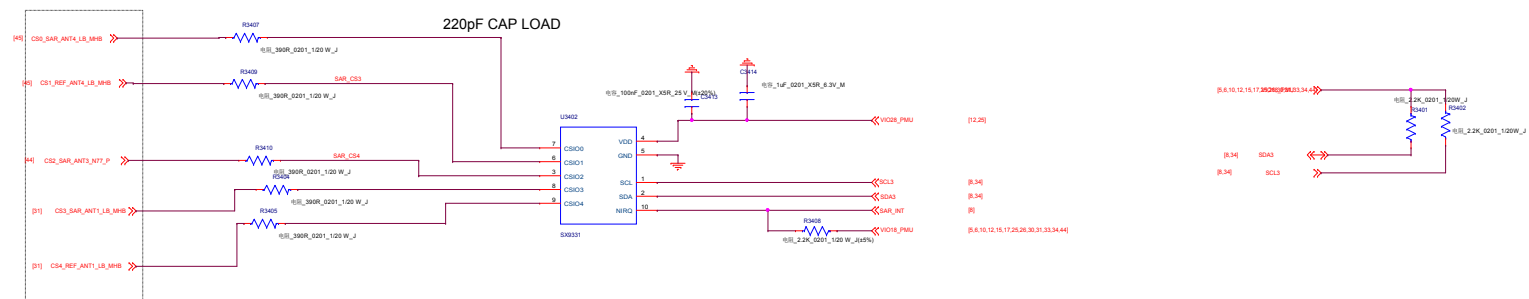
华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title 32.TEST_POINT/SHIELD		
Size D	Project <Doc>	Rev v1
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RF_NFC_not support



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		RF_NFC	
Size	Project	<Doc>	Rev
E			V1
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SAR_sensor Not support



Design notice:

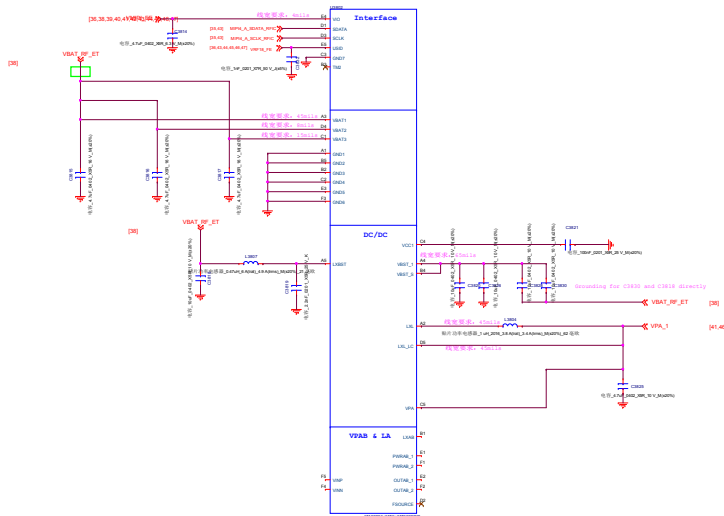
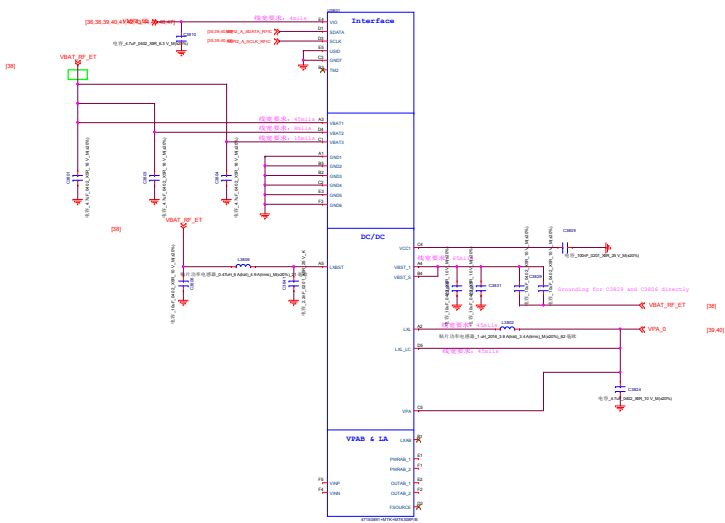
- 1) CS0_1 nets are differential pairs;
- 2) The width for each line, should be less than 0.1mm;
- 3) 0.2mm should be guaranteed between the ground and differential pairs;
- 4) And also, the distance between the differential line should be more than 0.2mm;
- 5) All the CSx nets should be protected by surrounding GND;

华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title SAR_sensor			
Size D	Project <Doc>		Rev v1
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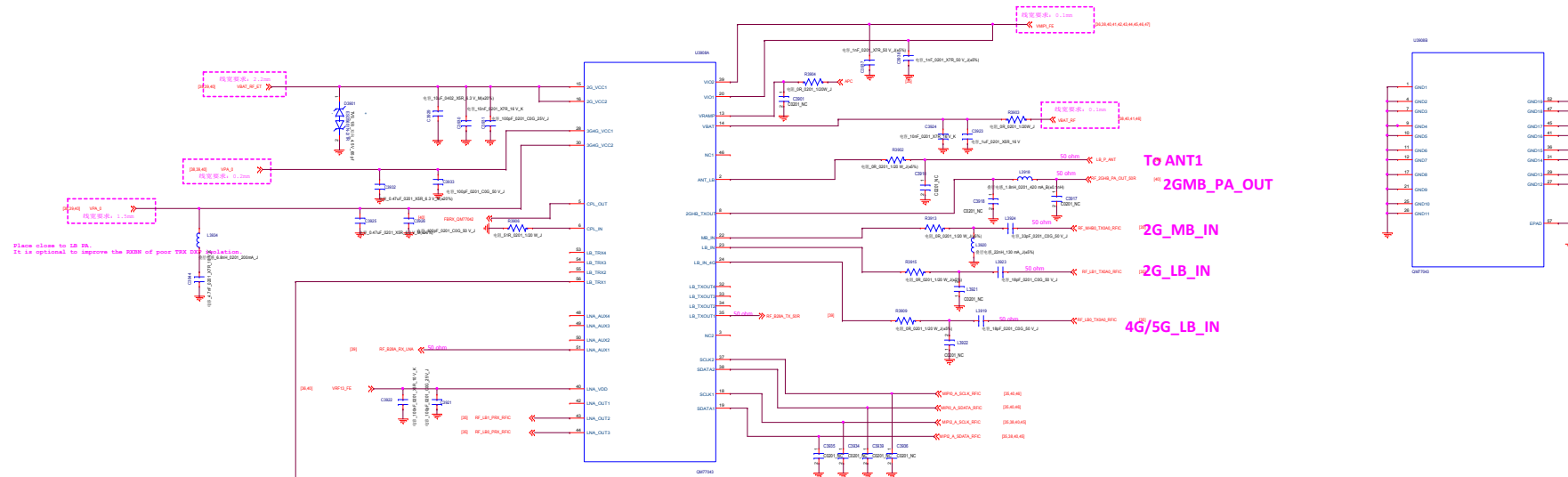
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		NC_FBRX	
Size D	Project <Doc>		Rev v1
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QM77042 and QM77048 for one APT

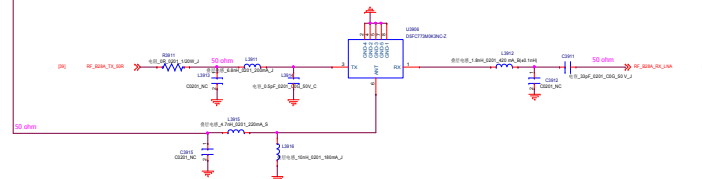
VC7643 and QM78200 for the other APT



2G_3G/4G/5G LB PA_QM77042



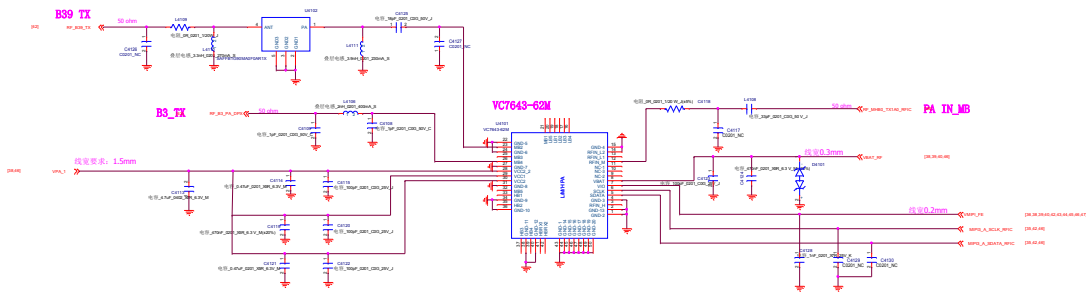
QM77042同QM77043 pin to pin



NTC温度检测
见P10的R1001 and NTC1001

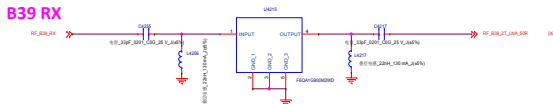
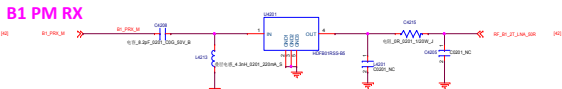
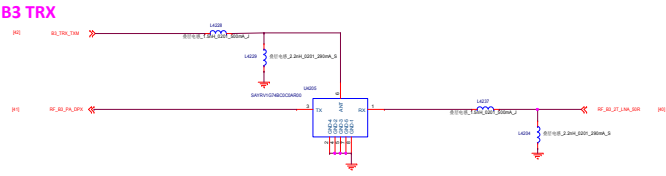
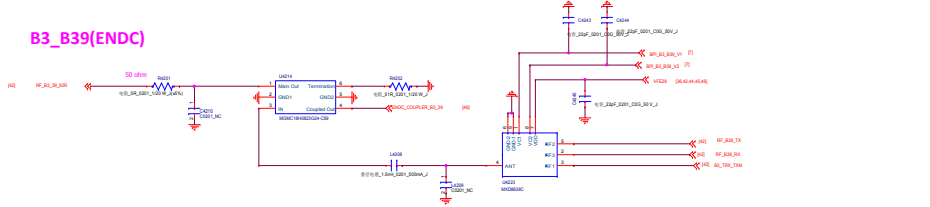
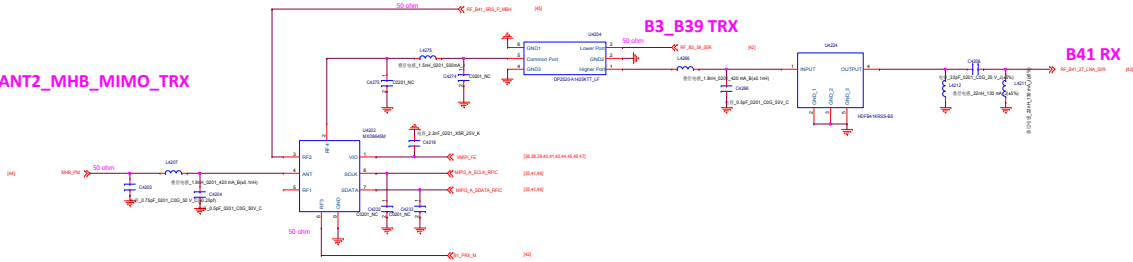
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title QM77042			
Size AQ	Project <Doc>	Rev v1	
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MMMB PA_MB ENDC



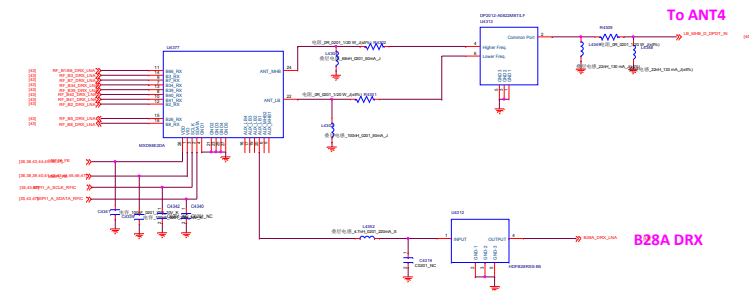
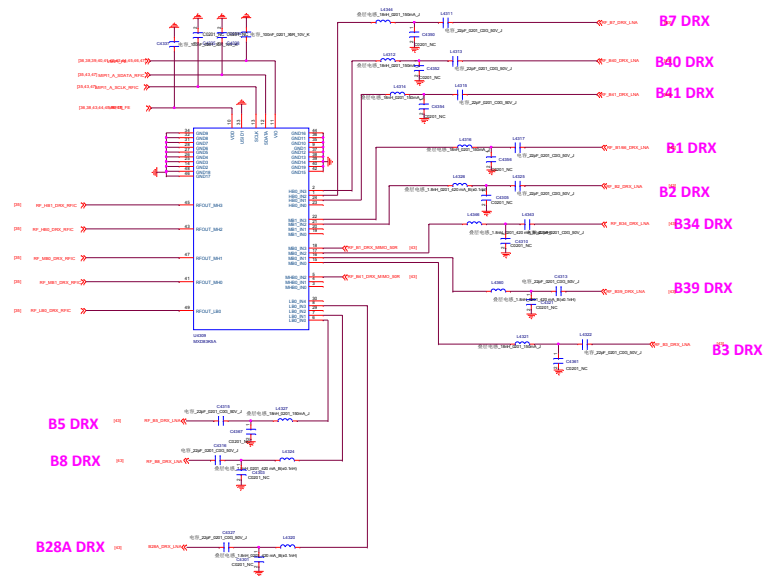
华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title		MMMB PA_MB ENDC	
Size	Project	<Doc>	Rev
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MMMB PA_VC7643_ENDC_MHB_TRX

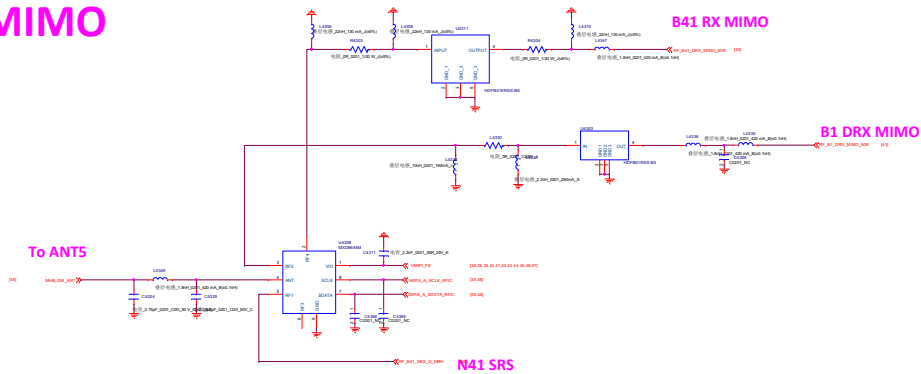


华勤通讯 Huaqin Telecom Technology Com.,Ltd		
Title MMMB PA_VC7643_ENDC_MHB_TRX		
Size A0	Project <Doc>	Rev v1
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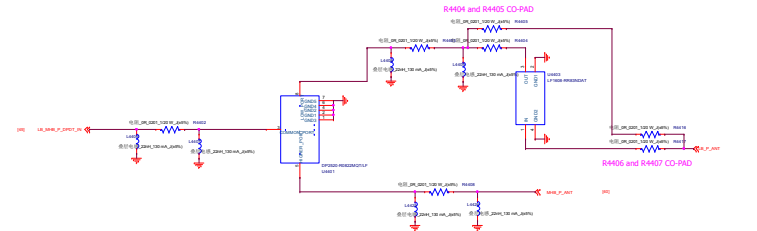
DRX_MXD98E2DA+MXD83K5A



DRX_MIMO

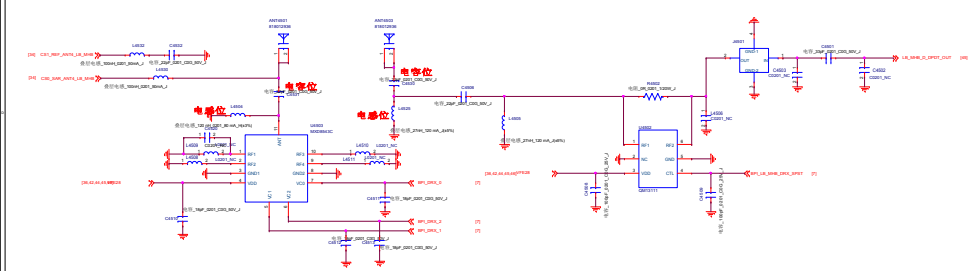


华勤通讯 Huaqin Telecom Technology Com.,Ltd			
Title MXD98E2DA_MXD83K5A			
Size E	Project <Doc>		Rev V1
Date: Tuesday, January 12, 2021	Sheet 43 of 48		

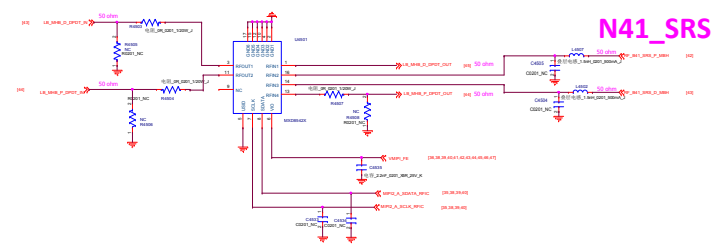
[illegible]

华勤通讯 Huaqin Telecom Technology Co.,Ltd			
Title ANT_P_PM			
Size E	Project <Doc>		Rev v1
Date: Tuesday, January 12, 2021		Sheet 44 of 48	

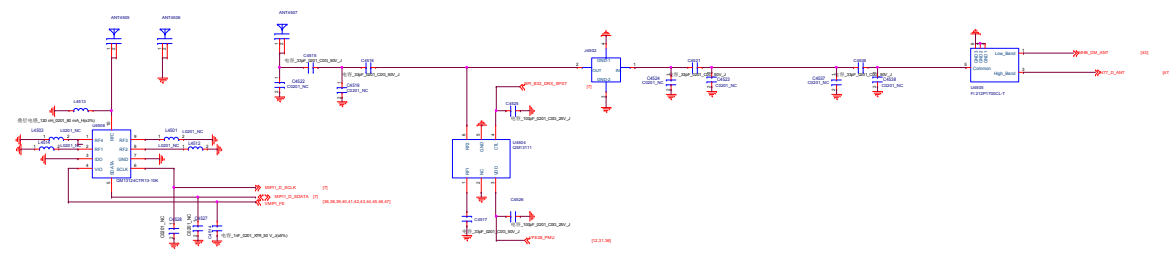
ANT4 (LB_MHB_D)



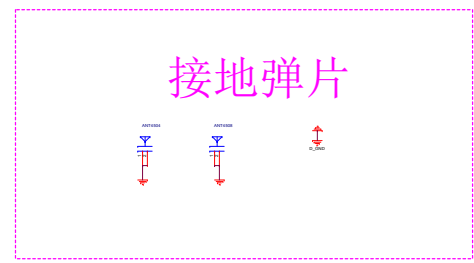
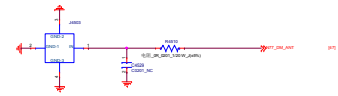
AsDIV_2P4T



ANT5 (MHB_DM/N77_D)



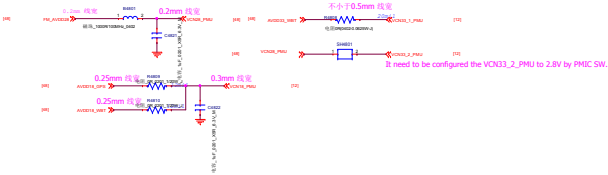
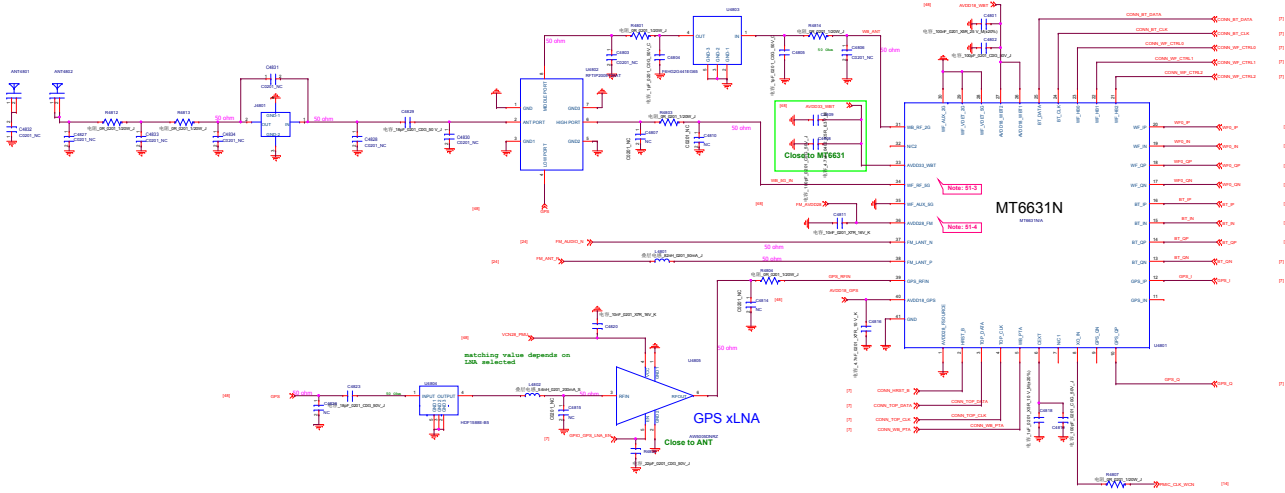
ANT6 (N77_DM) to 小板



华勤通讯 Huaqin Telecom Technology Com.,Ltd			
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ANT7
WiFi 2.4G & 5G_GPS

MT6631



Schematic design notice of "S1_CONNECTIVITY_CONSYS_MT6631"

Note 51-1: For RS015 size, please select 0402 size or larger one

Note 51-2: Please refer to MT6765 baseband design notice for VCN33_LDO selection guide

Note 51-3: If WiFi 5G not support, connect pin 34(WF_RF_5G) to GND

Note 51-4: Pin 36 (AVDD28_FM) must be connected to VCN28 even if FM not support